MMFormer: Macro–Micro Transformer for Small-Sample Classification of Mars Hyperspectral Image

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***Abstract*—Mars hyperspectral image classification (Mars HSIC) provides invaluable information for accurate mineral identification and Martian geology studies. However, this task remains challenging owing to scarce labeled samples, high spectral similarity, and environmental noise. Capturing both local details and global contexts is crucial for this task, yet convolutional neural networks (CNNs) struggle with global dependencies, while Transformers tend to neglect fine local details and incur unnecessary computational burden. To address this issue, we propose MMFormer, a macro–micro Transformer tailored for Mars HSIC. With a dual-branch architecture, our method adopts the *macro-kernel perception and micro-kernel modulation* strategy to synergize global contexts with local details in both spatial and spectral domains. In particular, the spatial branch adopts macro-kernel perception to capture long-range spatial dependencies, accompanied by micro-kernel modulation for local detail extraction. Meanwhile, the spectral branch utilizes an analogous strategy to perceive long-range spectral correlations and modulate fine-grained spectral variations simultaneously. To mitigate severe background interference, we introduce a center-prompt attention (CPA) module that injects a center-focused prompt to adaptively reweight attention toward the central pixel. To integrate these distinct feature representations, a cross-weighted fusion (CWF) module is constructed to dynamically fuse complementary spatial and spectral information. Extensive experiments on three Mars hyperspectral datasets demonstrate that MMFormer achieves strong performance and robustness under limited labeled samples and complex environmental conditions. The code is available at https://github.com/tingruifeng/TGRS_2026_MMFormer.**

***Index Terms*—Macro–micro attention (MMA), Mars hyperspectral image classification (Mars HSIC), small-sample learning, Transformer.**

I. Introduction

MARS hyperspectral image classification (Mars HSIC) plays a pivotal role in planetary science and deep space exploration [1], [2]. Accurate classification of Martian minerals provides indispensable scientific insights into the planet's geological evolution, past water activity, and atmospheric dynamics [3], [4]. Leveraging hundreds of contiguous and narrow spectral bands, hyperspectral images (HSIs) capture distinct spatial textures and fine spectral signatures, thereby facilitating the discrimination of diverse Martian mineral compositions [5], [6]. Consequently, driven by rapid advancements in deep-space exploration technology, Mars HSIC has recently attracted considerable scholarly attention [7], [8].

Traditional hyperspectral image classification (HSIC) methods, including spectral angle mapper (SAM) [9], Euclidean distance (ED) [10], and K-nearest neighbors (KNNs) [11], are predominantly based on spectral similarity matching. These approaches rely on handcrafted distance metrics while neglecting spatial contextual information. On the contrary, driven by advances in machine learning, data-driven techniques such as support vector machines (SVMs) [12], [13], random forest (RF) [14], and extreme learning machines (ELMs) [2] have been widely adopted for HSIC. Although they automatically learn feature representations through statistical frameworks, they still fail to effectively model the intricate spatial–spectral coupling relationships inherent in hyperspectral data.

As deep learning advances, convolutional neural networks (CNNs) have become a mainstream paradigm for extracting discriminative spatial and spectral features [15], [16]. Early methods primarily utilized 2-D convolutions to capture spatial textures [17]. Subsequently, to better integrate spatial and spectral information, 3-D convolutions and hybrid 2D–3D CNN frameworks were developed, including multichannel hybrid architectures designed for small-sample scenarios [18], [19], [20]. Furthermore, depthwise and grouped convolutions have been employed to reduce complexity [21], while attention-enhanced frameworks integrate spectral–spatial attention mechanisms [22]. For pixel-level classification, methods like CVSSN emphasize central pixel-neighborhood relationships [23], and some works integrate CNNs with graph neural networks (GNNs) to enhance global feature modeling [24]. However, the inherent locality of convolutional operations makes it difficult to effectively capture global contextual information and long-range dependencies.

To address the inherent limitation of CNNs in modeling long-range dependencies, Transformer-based models have emerged as a dominant architecture in computer vision

Received 4 February 2026; revised 22 April 2026; accepted 19 May 2026. Date of publication 25 May 2026; date of current version 12 June 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62371076 and Grant 62501087 and in part by the New Chongqing Youth Innovative Talents Project under Grant CSTB2024NSCQ-QCXMX0071. *(Corresponding author: Fulin Luo.)*

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Digital Object Identifier 10.1109/TGRS.2026.3696892

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[25], [26], [27]. Leveraging multi-head self-attention (MHSA) mechanisms [28], these models excel at modeling long-range spatial and spectral correlations, offering a powerful solution for HSIC. For global context modeling, representative approaches include HSI-BERT, which introduces a BERT-like pretraining paradigm [29], and SpectralFormer and SFTN, which employ spectral attention mechanisms [30], [31]. In terms of multiscale feature extraction, DSFormer utilizes dynamic sparse attention [32], DBMST adopts channel shrink modules [33], and MSTSENet and MsET integrate multiscale CNNs with attention mechanisms [34], [35]. Regarding positional encoding improvements, NG-CRPTN and HyperRoPE-SST employ relative position modeling and rotary embeddings, respectively [36], [37]. To address lightweight design and small-sample scenarios, GSC-ViT and PCET leverage groupwise separable convolutions and patch confidence enhancement [38], [39]. Furthermore, enhancements are observed in GAHT for global–local context balance [40], MASSFormer for cross-scale information mixing [41], and MorphFormer for structural feature capture [42]. However, pure Transformer architectures still struggle to efficiently extract fine-grained local spatial–spectral details critical for distinguishing similar types in HSIC.

To synergize the complementary strengths of CNNs and Transformers, hybrid architectures have been developed. In particular, SSFTT and SSRN leverage 3-D convolutions to extract spectral–spatial tokens [43], [44], while HiT deeply integrates convolutional operations into the Transformer backbone [45]. In terms of spectral–spatial fusion, SSCHFormer and synergistic CNN-Transformer networks combine spectral pyramid convolution with attention fusion [46], [47], whereas TNCCA employs CNN-enhanced cross-attention for multiscale features [48]. Advanced fusion mechanisms include CAF-Former for bridging local–global gaps [49], ISSFormer for parallel spectral–spatial branches [50], and CTFN for graph-encoded multiscale fusion [51]. Regarding small-sample and large-scale scenarios, CTA-net and PatchOut tackle limited training data and efficient segmentation, respectively [52], [53]. Furthermore, SAM-CTMapper proposes scale-aware mixed frameworks [54]. Collectively, these hybrid architectures have demonstrated excellent performance in terrestrial HSIC.

The aforementioned methods are primarily designed for terrestrial HSI analysis. However, directly applying these Earth-optimized models to Martian data presents significant challenges due to pronounced distributional differences. These differences primarily manifest as extreme label scarcity, severe atmospheric interference, and high spectral similarity among minerals. In particular, the issue of label scarcity is acute. Definitive mineral annotations rely primarily on limited in situ rover observations and expert visual analysis [55]. Orbital coverage limitations and high-mission costs further restrict the scale of usable labeled data [56]. Martian hyperspectral data also suffer from severe atmospheric interference. Here, fine dust and seasonal CO_2 strongly scatter and absorb light, introducing pronounced absorption features and reducing surface spectral contrast [57], [58]. High spectral similarity further complicates identification, with distinct mineral classes often exhibiting subtle, overlapping spectral features in the visible-near-infrared range [59]. Consequently, effective Mars HSIC demands methods that are robust to strong noise, capable of discerning subtle spectral differences, and effective under extreme scarcity of labeled samples. Recent Mars-specific studies have made significant progress in addressing label scarcity and feature learning. For instance, MCTGCL integrates CNN and Transformer branches and adopts graph contrastive learning to enhance feature representation under scarce labeled samples [1], while few-shot frameworks such as FSCILPR are constructed for incremental classification to sustain stable cross-task knowledge transfer [8]. Such methods have achieved remarkable progress in tackling limited labeled data and knowledge generalization challenges. Nevertheless, most of these approaches primarily focus on global feature modeling and task-specific adaptation. Capturing both local details and global contexts is crucial for this task. Local spatial information supports fine-grained material characterization, while global contextual dependencies facilitate overall regional perception, and the synergistic exploitation of both information is beneficial to comprehensive feature expression. To fulfill such multiscale feature modeling requirements, it is necessary to further conduct integrated extraction of local and global information.

Inspired by the "See Large, Focus Small" concept [60] and human visual perception mechanisms where global context perception guides the identification of focused targets, we propose a novel macro–micro Transformer (MMFormer) framework tailored for Mars HSIC. This framework integrates dual-branch spatial and spectral MMFormer modules, operating under a *macro-kernel perception and micro-kernel modulation* strategy. In particular, the spatial MMFormer captures multiscale macro spatial contexts to refine local micro spatial details, while the spectral MMFormer exploits long-range macro spectral dependencies to enhance discriminative micro spectral feature representations. Furthermore, recognizing that central pixels carry the most critical identity information, we introduce a center-prompt attention (CPA) module to emphasize the central target and mitigate noise via adaptive reweighting. Finally, a cross-weighted fusion (CWF) module is designed to dynamically synthesize the extracted spatial and spectral features.

The main contributions of our proposed framework can be summarized as follows.

1) We propose the macro–micro attention (MMA) mechanism to realize effective extraction of features following a global-to-local sequence for Mars HSIC.
2) We design spatial MMFormer and spectral MMFormer based on MMA to separately conduct global-to-local feature extraction for spatial and spectral dimensions.
3) We design the CPA module to emphasize central pixel information and suppress pervasive noise.
4) We introduce the CWF module to dynamically fuse spatial and spectral features with cross-adaptive weights.

The remainder of the article is organized as follows. Section II details the design of our proposed model. Section III presents extensive experiments and analyzes the classification

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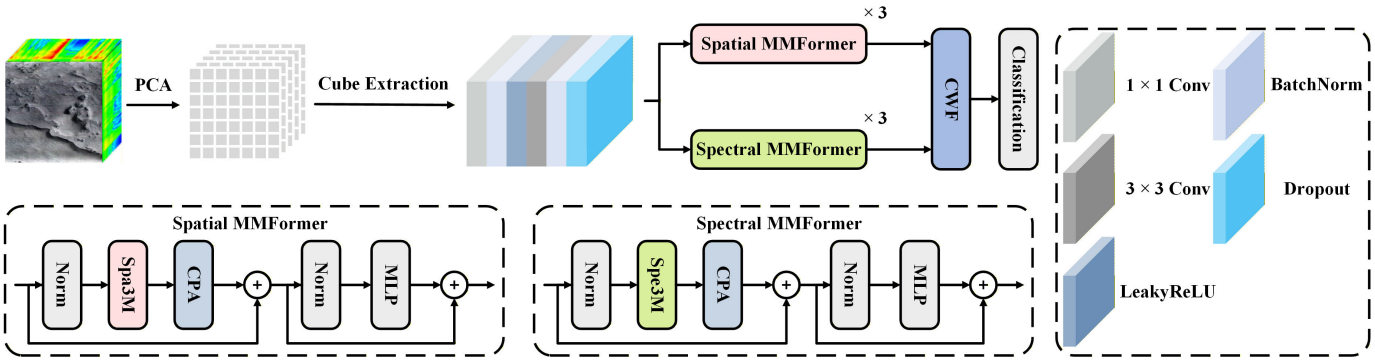


Fig. 1. Overall architecture of the proposed MMFormer framework. The raw hyperspectral data are first processed by PCA dimensionality reduction and then split into 3-D patch cubes. The patches are then fed into two parallel branches composed of three spatial MMFormer blocks and three spectral MMFormer blocks, respectively. The CPA module is embedded as an internal component within each MMFormer block. Finally, the multiscale spatial–spectral features are adaptively fused by the CWF module and fed into the classifier for label prediction.

results. Section IV presents the conclusions of this work and outlines potential directions for future research.

II. Proposed Method

A. Overview

The overall architecture of the proposed MMFormer framework is illustrated in Fig. 1. This framework comprises four synergistic components, each tailored to mitigate specific challenges in Mars HSIC. The model adopts two parallel branches: the spatial MMFormer branch for spatial feature learning and the spectral MMFormer branch for spectral dependency modeling. The CPA module is embedded as an internal component within each spatial and spectral MMFormer block to enhance the representation of central regions. The CWF module is further developed to dynamically integrate complementary spatial–spectral information.

The pipeline initiates with dimensionality reduction via principal component analysis (PCA) on the raw hyperspectral data, where the data are reduced to 30 principal components while preserving discriminative information. The dimension-reduced data are subsequently partitioned into overlapping 3-D patch cubes. These patches are then fed into two parallel branches composed of three spatial MMFormer blocks and three spectral MMFormer blocks, respectively. Finally, the outputs from both branches are adaptively fused by the CWF module and forwarded to a multilayer perceptron for classification.

B. Spatial MMFormer

The spatial MMFormer is designed to extract hierarchical spatial features from hyperspectral imagery. Traditional CNNs often struggle to capture global contextual dependencies due to their inherent locality, while pure Transformers tend to neglect fine-grained local details critical for distinguishing similar mineral types. To address these limitations, we propose the spatial MMFormer based on the innovative *macro-kernel perception and micro-kernel modulation* strategy. The complete module integrates the Spa3M block, batch normalization, the CPA module for central pixel representation, and an MLP layer, with residual connections employed throughout to ensure stable gradient flow.

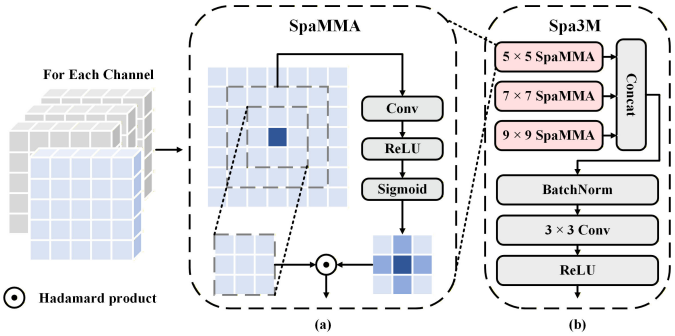


Fig. 2. Architecture of the Spa3M block. (a) SpaMMA mechanism and (b) Complete Spa3M block with multiscale feature integration. In (b), 5 × 5, 7 × 7, and 9 × 9 denote the spatial macro-kernel sizes of the three parallel branches.

The fundamental unit of the spatial MMFormer is the spatial macro–micro attention (SpaMMA) Block. As shown in Fig. 2(a), the SpaMMA Block implements a dynamic convolution mechanism where global contexts guide the refinement of local details. For an input feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, the macro-kernel perception first generates adaptive modulation weights. Unlike standard attention maps that often produce channelwise statistics, this branch outputs a spatial weight tensor $\mathbf{W} \in \mathbb{R}^{B\times (s_k^{spa})^2\times H\times W}$, representing the dynamic parameters for the local convolution kernel

$$\mathbf{W} = \sigma\left(\mathrm{ReLU}\left(\mathrm{Conv}_{l_k^{spa}\times l_k^{spa}}(\mathbf{X})\right)\right) \tag{1}$$

where l_k^{spa} denotes the spatial macro-kernel size. This weight tensor is spatially adaptive and channel shared. It encodes spatial structure priors derived from the large kernel, applying this guidance uniformly across all feature channels to ensure consistent feature extraction.

To apply these spatial weights, the local features are first restructured by unfolding $\mathbf{X}$ into a 5-D tensor $\mathbf{X}_{\mathrm{local}} \in \mathbb{R}^{B\times C\times (s_k^{spa})^2\times H\times W}$, where the $(s_k^{spa})^2$ dimension explicitly models the local spatial structure. The modulation is then

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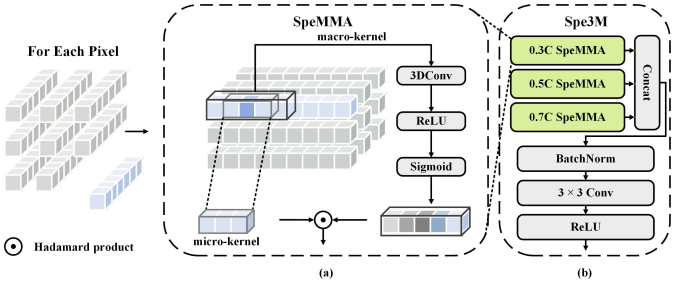


Fig. 3. Architecture of the Spe3M block. (a) SpeMMA branches and (b) complete the Spe3M block with multiscale integration. In (b), $0.3C$, $0.5C$, and $0.7C$ denote the spectral macro-kernel sizes of the three parallel branches, where C denotes the number of input channels to this block.

performed via a broadcasting mechanism

$$\mathbf{F}_{\text{spa}} = \text{Fold}\left(\text{Reshape}\left(\mathbf{X}_{\text{local}} \odot \mathbf{W}'\right)\right) \tag{2}$$

where $\mathbf{W}'$ is obtained by unsqueezing $\mathbf{W}$ to match the channel dimension C (i.e., $\mathbf{W}' \in \mathbb{R}^{B\times 1\times (s_k^{spa})^2\times H\times W}$). This broadcasting strategy allows the model to efficiently apply the globally perceived weights $\mathbf{W}$ to modulate the local structures of all C channels simultaneously. The modulated features are finally folded back to the original spatial resolution.

The spatial MMFormer extends this unit by integrating three parallel SpaMMA branches to model multiscale dependencies. As illustrated in Fig. 2(b), the Spa3M block orchestrates these branches with distinct macro-kernel sizes (typically $l_k^{spa} \in \{5, 7, 9\}$). Corresponding to the macro-kernels for global context modeling, the spatial micro-kernel size s_k^{spa} is set to 3 to capture fine-grained local structures while maintaining compact local feature modeling. This design achieves immediate multiscale perception within a single computational unit. The outputs of the three branches $\mathbf{F}_{\text{spa}}^1, \mathbf{F}_{\text{spa}}^2, \mathbf{F}_{\text{spa}}^3$ are concatenated and fused via batch normalization, a 3×3 convolution, and ReLU

$$\mathbf{F}_{\text{Spa3M}} = \text{ReLU}\left(\text{Conv}\left(\text{BN}\left(\text{Concat}\left(\mathbf{F}_{\text{spa}}^1, \mathbf{F}_{\text{spa}}^2, \mathbf{F}_{\text{spa}}^3\right)\right)\right)\right). \tag{3}$$

The integration of parallel multiscale processing enables the spatial MMFormer to effectively capture both long-range contextual dependencies and fine-grained local details. This architecture provides a robust solution for modeling complex spatial structures in hyperspectral imagery, particularly under the challenging conditions of small-sample Mars hyperspectral classification.

C. Spectral MMFormer

The spectral MMFormer serves as the complementary spectral branch dedicated to modeling multiscale dependencies across wavelength bands. As illustrated in Fig. 3, it sequentially comprises a normalization layer, a Spe3M block, a CPA block, and an MLP layer. Among these, the Spe3M block acts as the primary feature extractor, adopting a *macro-kernel perception and micro-kernel modulation* strategy to capture comprehensive spectral correlations.

The fundamental unit of the Spe3M block is the spectral MMA (SpeMMA) branch. As shown in Fig. 3(a), the SpeMMA mechanism jointly performs global spectral weighting and local spectral modulation. To capture long-range spectral dependencies, the input is first reshaped into a 5-D tensor $\mathbf{X}_{5\text{d}} \in \mathbb{R}^{B\times 1\times C_{\text{in}}\times H\times W}$ by inserting a singleton dimension. We apply a 3-D convolution with a *spectral macro-kernel* of size $(l_k^{spe}, 1, 1)$ to this tensor. This convolution slides exclusively along the spectral dimension while maintaining a 1×1 spatial footprint, generating a spatially adaptive weight tensor $\mathbf{W}_{\text{spe}}$ after permutation

$$\mathbf{W}_{\text{spe}} = \sigma\left(\text{ReLU}\left(\text{Conv3D}_{(l_k^{spe},1,1)}\left(\mathbf{X}_{5\text{d}}\right)\right)\right) \tag{4}$$

where l_k^{spe} denotes the spectral macro-kernel size determining the receptive field along the channel dimension. The 3-D convolution uses s_k^{spe} output channels, and the generated weight tensor $\mathbf{W}_{\text{spe}} \in \mathbb{R}^{B\times C_{\text{in}}\times s_k^{spe}\times H\times W}$ serves as the dynamic parameters for local spectral refinement, where s_k^{spe} represents the spectral micro-kernel size.

In parallel with the weight generation, the input is permuted to $\mathbf{X}_{\text{perm}} \in \mathbb{R}^{B\times H\times W\times C_{\text{in}}}$ to align the spectral dimension to the last axis. The permuted feature is then unfolded with the fixed micro-kernel size $s_k^{spe} = 9$, producing local spectral structures $\mathbf{X}_{\text{local}} \in \mathbb{R}^{B\times C_{\text{in}}\times s_k^{spe}\times H\times W}$

$$\mathbf{X}_{\text{local}} = \text{Unfold}_{s_k^{spe}}\left(\mathbf{X}_{\text{perm}}\right). \tag{5}$$

The spectral patches are adaptively modulated by the precomputed weight tensor via elementwise multiplication, followed by summation over the micro-kernel dimension to produce the final output feature $\mathbf{F}_{\text{spe}} \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$

$$\mathbf{F}_{\text{spe}}(c, h, w) = \sum_{i=1}^{s_k^{spe}} \left(\mathbf{X}_{\text{local}}(c, i, h, w) \odot \mathbf{W}_{\text{spe}}(c, i, h, w)\right) \tag{6}$$

where i indexes the local window of the spectral micro-kernel.

The Spe3M block extends this mechanism by integrating three parallel SpeMMA branches to model multiscale dependencies. As shown in Fig. 3(b), these branches employ distinct macro-kernel sizes determined by proportional scaling ratios $\{0.3, 0.5, 0.7\}$, where $l_k^{spe} = \text{Round}(C_{\text{in}} \times \text{ratio})$ is enforced to be odd. The outputs $\mathbf{F}_{\text{spe}}^1, \mathbf{F}_{\text{spe}}^2$, and $\mathbf{F}_{\text{spe}}^3$ are concatenated and fused via BatchNorm, a 3×3 convolution, and ReLU

$$\mathbf{F}_{\text{Spe3M}} = \text{ReLU}\left(\text{Conv}\left(\text{BN}\left(\text{Concat}\left(\mathbf{F}_{\text{spe}}^1, \mathbf{F}_{\text{spe}}^2, \mathbf{F}_{\text{spe}}^3\right)\right)\right)\right). \tag{7}$$

The integration of parallel multiscale spectral processing enables the spectral MMFormer to effectively capture both long-range spectral dependencies and fine-grained local spectral signatures. This architecture provides a robust solution for modeling complex spectral structures in hyperspectral imagery, particularly under the challenging conditions of small-sample Mars hyperspectral classification, where spectral similarity among different mineral types poses significant classification challenges.

D. Center Prompt Attention

CPA is designed for pixel-level HSIC tasks in Fig. 4 where accurate identification of the central pixel within each patch is critical. Traditional attention mechanisms typically treat all spatial positions uniformly, which can dilute the

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discriminative power of the central region that serves as the primary classification target. CPA addresses this limitation through a dual-path architecture that combines channelwise and spatial enhancement strategies. This design amplifies central pixel features while maintaining contextual information from surrounding regions, establishing an effective approach for HSI analysis.

The CPA module processes an input feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times K\times K}$, where $K\times K$ represents the patch size. The center-biased channel excitation pathway employs a *Center-Biased Average Pooling* strategy. Unlike standard global pooling that treats all positions equally, we construct a spatial weighting matrix $\mathbf{W}_{\text{center}}$ where the central pixel is assigned a dominant proportion α of the total weight while the remaining weight $1-\alpha$ is uniformly distributed among the surrounding context pixels

$$\mathbf{W}_{\text{center}}(i,j) = \begin{cases} \alpha, & \text{if } (i,j) = (c_x, c_y) \\ \dfrac{1-\alpha}{K^2-1}, & \text{otherwise} \end{cases} \tag{8}$$

where $\alpha \in (0,1)$ controls the intensity of the center bias (set to 0.5 in our experiments) and (c_x, c_y) denotes the coordinates of the central pixel. This weighting matrix satisfies $\sum_{i,j} \mathbf{W}_{\text{center}}(i,j) = 1$. The pooled channel descriptor $\mathbf{z}$ is then computed via elementwise multiplication and summation

$$\mathbf{z} = \sum_{i=1}^{K}\sum_{j=1}^{K} \mathbf{X}_{:,:,i,j} \odot \mathbf{W}_{\text{center}}(i,j). \tag{9}$$

The descriptor is processed through a channel excitation network that captures spectral relationships

$$\mathbf{w}_c = \sigma\left(\text{Linear}_2\left(\text{ReLU}\left(\text{Linear}_1\left(\mathbf{z}\right)\right)\right)\right). \tag{10}$$

The resulting channel weights $\mathbf{w}_c$ modulate the input features through elementwise multiplication: $\mathbf{X}_c = \mathbf{X} \odot \mathbf{w}_c$.

The spatial Gaussian prompting branch employs an attention mask that prioritizes regions near the patch center. This spatial prioritization uses a Gaussian distribution

$$\mathbf{M}(i,j) = \frac{\exp\left(-\dfrac{(i-c_x)^2 + (j-c_y)^2}{2\sigma^2}\right)}{\max\left(\exp\left(-\dfrac{(i-c_x)^2 + (j-c_y)^2}{2\sigma^2}\right)\right)}. \tag{11}$$

This approach creates an attention gradient that emphasizes the central pixel while incorporating contextual information. The spatially refined features are obtained as $\mathbf{X}_s = \mathbf{X} \odot \mathbf{M}$.

The fusion process combines both enhancement pathways

$$\mathbf{X}_{\text{cpa}} = \text{ReLU}\left(\text{Conv}_{1\times 1}\left(\text{Concat}\left(\mathbf{X}_c, \mathbf{X}_s\right)\right)\right). \tag{12}$$

This integrated approach ensures the enhanced features retain the discriminative power of both strategies. The CPA module enhances the discriminative ability of central pixels while retaining contextual information, which is particularly important for Mars HSI classification tasks where labeled samples are scarce and similar mineral pixels are easily confused. This design provides robust feature support for subsequent classification by highlighting the core target pixels and ensuring diagnostically valuable information receives appropriate emphasis.

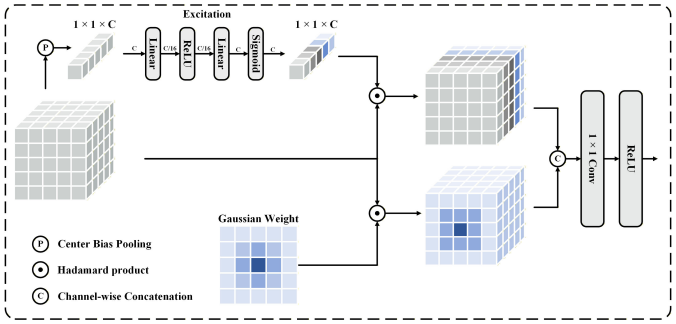


Fig. 4. Overview of the CPA module. The architecture utilizes a dual-path strategy combining channelwise excitation and spatial center-bias pooling to enhance the discriminative features of the central pixel.

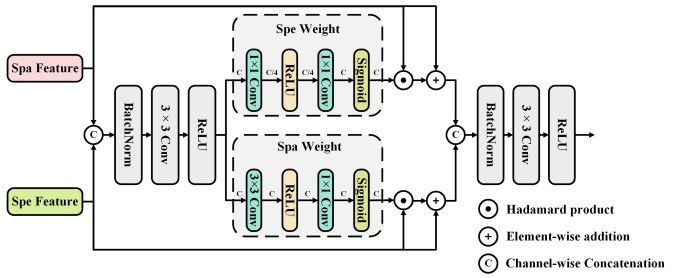


Fig. 5. Architecture of the proposed CWF module.

E. Cross Weighted Fusion

CWF is designed to enable bidirectional modulation between spatial and spectral features, facilitating adaptive interaction across domains. Unlike conventional fusion methods that treat spatial and spectral information as static components, CWF generates adaptive cross-domain weights to dynamically enhance complementary cues from both branches, ensuring coherent and task-relevant feature integration. The architecture of CWF is illustrated in Fig. 5.

The fusion process begins with prefusion feature aggregation. Let $\mathbf{F}_{\text{spa}}$ and $\mathbf{F}_{\text{spe}}$ denote the spatial and spectral input features, respectively. They are first concatenated along the channel dimension to form a combined representation $\mathbf{X}_{\text{cat}} \in \mathbb{R}^{B\times 2C\times H\times W}$. This tensor undergoes batch normalization (BN) to stabilize training, followed by a 3×3 convolution to compress the channel dimension back to C and a ReLU activation to introduce nonlinearity. This step yields a preliminary aggregated feature $\mathbf{X}_{\text{pre}}$ that captures basic cross-domain interactions

$$\mathbf{X}_{\text{pre}} = \text{ReLU}\left(\text{Conv}_{3\times 3}\left(\text{BN}\left(\text{Concat}\left(\mathbf{F}_{\text{spa}}, \mathbf{F}_{\text{spe}}\right)\right)\right)\right). \tag{13}$$

From $\mathbf{X}_{\text{pre}}$, two parallel weight generators produce adaptive weights tailored for bidirectional modulation. The spatial weight generator employs a 3×3 convolution (to encode local spatial context) followed by a ReLU, a 1×1 convolution (to adjust channel dimensions), and finally a sigmoid function to normalize weights to the $[0,1]$ range, resulting in $\mathbf{W}_{\text{spa}}$. On the contrary, the spectral weight generator uses two 1×1 convolutions with a channel reduction ratio of 4 (to focus on channelwise spectral interactions), interspersed with

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ReLU, ending with sigmoid normalization to produce $\mathbf{W}_{\text{spe}}$. Mathematically, these weights are defined as

$$\mathbf{W}_{\text{spa}} = \sigma\left(\text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{Conv}_{3\times3}\left(\mathbf{X}_{\text{pre}}\right)\right)\right)\right) \tag{14}$$

$$\mathbf{W}_{\text{spe}} = \sigma\left(\text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{Conv}_{1\times1}\left(\mathbf{X}_{\text{pre}}\right)\right)\right)\right). \tag{15}$$

Both $\mathbf{W}_{\text{spa}}$ and $\mathbf{W}_{\text{spe}}$ match the dimensions of the input features ($B \times C \times H \times W$), enabling element-wise modulation.

These cross-domain weights drive bidirectional feature refinement with residual connections to preserve the original feature integrity. In particular, spatial features $\mathbf{F}_{\text{spa}}$ are modulated by spectral weights $\mathbf{W}_{\text{spe}}$ to incorporate spectral discriminative cues, while spectral features $\mathbf{F}_{\text{spe}}$ are enriched by spatial weights $\mathbf{W}_{\text{spa}}$ to integrate spatial contextual information. The residual terms ensure that core feature information is retained during modulation

$$\begin{cases}\mathbf{F}_{\text{spa}}^{\text{mod}} = \mathbf{F}_{\text{spa}} \odot \mathbf{W}_{\text{spe}} + \mathbf{F}_{\text{spa}} \\ \mathbf{F}_{\text{spe}}^{\text{mod}} = \mathbf{F}_{\text{spe}} \odot \mathbf{W}_{\text{spa}} + \mathbf{F}_{\text{spe}}.\end{cases} \tag{16}$$

The modulated features ($\mathbf{F}_{\text{spa}}^{\text{mod}}$ and $\mathbf{F}_{\text{spe}}^{\text{mod}}$) are then concatenated again, undergoing BN, a 3×3 convolution, and ReLU to produce the final fused feature $\mathbf{F}_{\text{fusion}}$. This step aggregates the refined cross-domain information into a unified representation

$$\mathbf{F}_{\text{fusion}} = \text{ReLU}\left(\text{Conv}_{3\times3}\left(\text{BN}\left(\text{Concat}\left(\mathbf{F}_{\text{spa}}^{\text{mod}}, \mathbf{F}_{\text{spe}}^{\text{mod}}\right)\right)\right)\right). \tag{17}$$

This comprehensive fusion strategy ensures that the model leverages both spatial context and spectral signatures in a dynamically adaptive manner.

III. Experiment and Analysis

A. Data Description

To validate the proposed Mars hyperspectral classification method, the public HyMars dataset [61] was adopted, which is specifically designed for mineral classification tasks on the Martian surface. This dataset comprises three subdatasets corresponding to typical Martian geological regions, namely Holden crater (HC), Nili Fossae (NF), and Utopia Planitia (UP). All subdatasets are derived from HSIs acquired by the compact reconnaissance imaging spectrometer for Mars (CRISM) onboard the Mars reconnaissance orbiter (MRO). With a spatial resolution of 18 m/pixel and a spectral range of 0.36–3.93 μm, the CRISM data provide robust support for Martian mineral identification.

The HC dataset corresponds to the fluviolacustrine geological region of HC, with an image size of 418×595 pixels and 440 valid spectral bands. The NF dataset is sourced from NF, a region indicative of potential hydrothermal activity, featuring an image size of 478×593 pixels and 425 valid bands. The UP dataset is located at the landing site of China's Tianwen-1 rover "Zhurong," with an image size of 478×595 pixels and 432 valid bands. To simulate the small-sample scenario of Mars hyperspectral classification, each mineral class is assigned only ten labeled training samples, resulting in 60, 90, and 90 training samples for the HC, NF, and UP datasets, respectively, with the remaining samples allocated to the test set. Detailed information is provided in Tables I–III.

TABLE I
Category Names and Sample Distribution in the HC Dataset

Class ID	Color	Class Name	Train	Test	Total
1		Analcime	10	930	940
2		Plagioclase	10	1462	1472
3		Prehnite	10	1550	1560
4		High-Ca Pyroxene	10	6953	6963
5		Serpentine	10	8687	8697
6		Margarite	10	448	458
		Total	60	20030	20090

TABLE II
Category Names and Sample Distribution in the NF Dataset

Class ID	Color	Class Name	Train	Test	Total
1		Fe-Olivine	10	8553	8563
2		Epidote	10	986	996
3		Chlorite	10	376	386
4		Bassanite	10	3276	3286
5		Illite/Muscovite	10	218	228
6		Mg-Carbonate	10	4857	4867
7		Plagioclase	10	1047	1057
8		Prehnite	10	1151	1161
9		Serpentine	10	6156	6166
		Total	90	26620	26710

TABLE III
Category Names and Sample Distribution in the UP Dataset

Class ID	Color	Class Name	Train	Test	Total
1		Analcime	10	1741	1751
2		Bassanite	10	5797	5807
3		High-Ca Pyroxene	10	422	432
4		Illite/Muscovite	10	262	272
5		Low-Ca Pyroxene	10	6764	6774
6		Mg-Smectite	10	1156	1166
7		Monohydrated sulfate	10	330	340
8		Plagioclase	10	192	202
9		Prehnite	10	584	594
		Total	90	17248	17338

B. Experimental Settings

1) Evaluation Indicators: To quantitatively evaluate the classification performance of the proposed method, three widely recognized indicators in HSIC were adopted: overall accuracy (OA), average accuracy (AA), and Kappa Coefficient (κ). OA denotes the proportion of correctly classified samples in the total test samples, reflecting the overall classification effectiveness. AA is the average value of classification accuracy across all mineral classes, which mitigates the impact of uneven sample distribution on evaluation results. κ quantifies the consistency between predicted outcomes and ground truth, with a value range of $[-1, 1]$, and higher values indicate more reliable classification results. All indicators were calculated based on the confusion matrix of the test set, with results retained to four decimal places.

2) Configuration: All experiments were conducted on a unified hardware and software platform to guarantee reproducibility. The hardware environment was equipped with an NVIDIA GeForce RTX 3090 GPU, paired with NVIDIA driver

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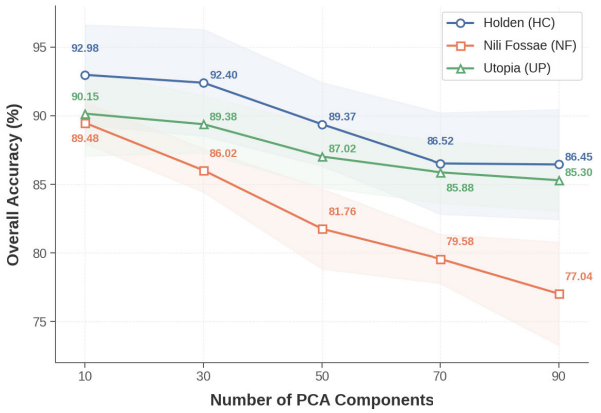


Fig. 6. OA with standard deviation on the HC, NF, and UP datasets for different numbers of PCA components.

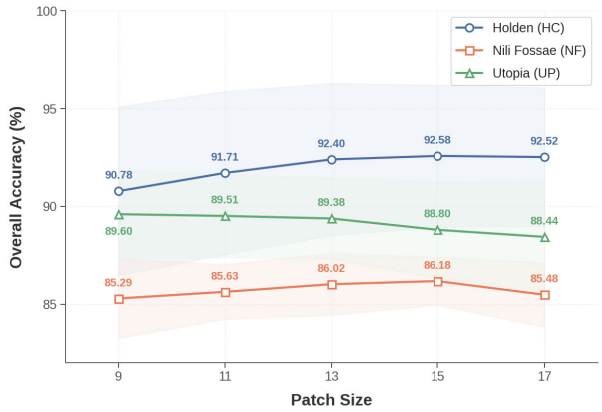


Fig. 7. OA with standard deviation on the HC, NF, and UP datasets for different patch sizes.

version 470.141.03 and CUDA 11.4 to support efficient parallel computing. The software environment was built on a Linux operating system, utilizing Python 3.8 as the programming language and PyTorch as the deep-learning framework.

To ensure the stability and robustness of the results, the experiment was repeated ten times with distinct random seeds starting from 42 (incremented by 1 for each run). For model training, the training set was constructed with ten labeled samples per class, and the remaining samples were allocated to the test set. The Adam optimizer was employed with an initial learning rate of 1×10^{-4}, a batch size of 128, and a maximum of 100 training epochs. The cross-entropy loss function was used for classification optimization, and a dropout layer with a probability of 0.1 was incorporated to alleviate overfitting. All compared methods uniformly adopted a patch size of 13×13 and 30-D PCA preprocessing to maintain consistent experimental conditions.

C. Sensitivity Analysis

To verify the robustness of the proposed MMFormer and systematically evaluate the sensitivity of key experimental parameters, comprehensive comparative experiments were conducted. This analysis focuses on five critical factors that influence spatial–spectral feature extraction: the number of PCA components for dimensionality reduction, the spatial patch size, the spectral micro-kernel size (s_k^{spe}), the center-bias weight α, and the Gaussian prompt σ. OA was adopted as the primary evaluation metric throughout this analysis.

1) PCA Components Sensitivity: PCA was employed as a preprocessing step to reduce the dimensionality of hyperspectral data while preserving the most informative spectral variance. The number of PCA components is a critical parameter that balances information retention against computational complexity and potential noise interference. Five configurations were evaluated: 10, 30, 50, 70, and 90 components, with results visualized in Fig. 6. The experimental results reveal that models with fewer PCA components (10 and 30) consistently outperform those with higher component counts across all datasets, indicating that excessive spectral dimensions can introduce noise and hinder feature learning. This phenomenon is particularly pronounced in Martian hyperspectral scenarios where the original data contain numerous spectral channels with varying levels of redundancy and atmospheric interference. Without proper dimensionality reduction, the high-dimensional input can lead to training instability and degrade model effectiveness due to the curse of dimensionality. Although ten components yield competitive accuracy, such a low dimensionality carries a risk of discarding subtle spectral features essential for distinguishing complex mineralogical signatures. On the contrary, 30 components provide a more robust tradeoff: retaining sufficient spectral information to capture diverse geological features while effectively filtering out noise. A clear performance degradation trend is observed as the number of components increases beyond 30, with OA on NF dropping from 86.02% (30 components) to 77.04% (90 components), representing a substantial decrease. Therefore, 30 PCA components were adopted for all subsequent experiments to ensure the reliability of feature representation.

2) Spatial Patch Size Sensitivity: The spatial patch size determines the spatial context scope around each central pixel. Five configurations were evaluated: 9×9, 11×11, 13×13, 15×15, and 17×17, with quantitative results visualized in Fig. 7. Across all three datasets, the classification performance exhibits excellent stability with only slight fluctuations: OA values range from 90.78% to 92.58% on HC, 85.29% to 86.18% on NF, and 88.44% to 89.60% on UP. This robustness indicates that the proposed model architecture, enhanced by the CPA mechanism and multiscale Spa3M module, can adapt well to variations in spatial context. The center-prompt mechanism prioritizes the features of the target pixel, while the multiscale attention compensates for contextual variations caused by patch size changes, thus ensuring reliable capture of spatial information regardless of the selected spatial scope. Smaller patches may lack sufficient contextual information for complex scenes, as seen in HC and NF, while larger patches tend to introduce redundant background noise as observed in UP. Therefore, the 13×13 patch size was selected for all subsequent experiments as it achieves the most balanced performance across all datasets.

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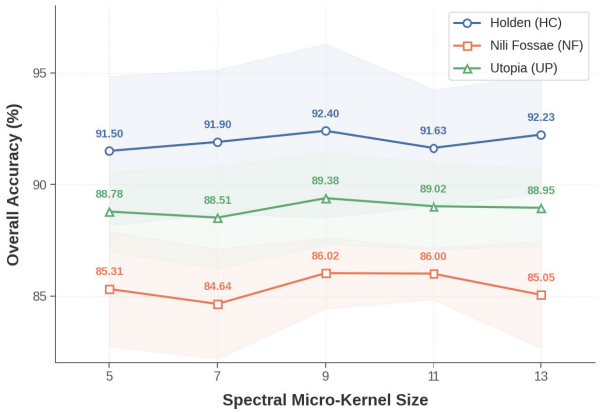


Fig. 8. OA with standard deviation on the HC, NF, and UP datasets for different spectral micro-kernel sizes (s_k^{spe}).

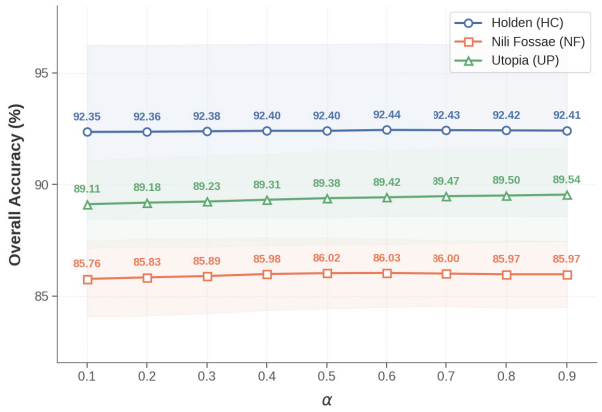


Fig. 9. OA with standard deviation on the HC, NF, and UP datasets for different center-bias weight α.

3) Spectral Micro-Kernel Size Sensitivity: The spectral micro-kernel size (s_k^{spe}) governs the range of wavelength dependencies modeled in the Spe3M block. Five settings were evaluated in the experiments: $s_k^{spe} = 5, 7, 9, 11, 13$, with detailed results presented in Fig. 8. Consistent with the analysis of spatial patch size, the model maintains high robustness across different s_k^{spe} values: OA values on HC fluctuate marginally from 91.50% to 92.40%, 84.64% to 86.02% on NF, and 88.51% to 89.38% on UP. In particular, the configuration with $s_k^{spe} = 9$ achieves the best overall performance across all datasets, yielding OA scores of 92.40% $\pm$ 3.89 on HC, 86.02%$\pm$1.59 on NF, and 89.38% $\pm$ 2.09 on UP. The observed insensitivity of the model to variations in the spectral kernel size is attributed to the innovative *macro-kernel perception and micro-kernel modulation* framework in the Spe3M block. While the macro-kernel captures flexible spectral ranges, the micro-kernel modulation dynamically adapts to local spectral correlations, thus reducing the model's reliance on fixed kernel sizes and enhancing its adaptability to the diverse spectral characteristics of different Martian geological formations. Therefore, $s_k^{spe} = 9$ was adopted for all subsequent experiments.

4) Center-Bias Weight α Sensitivity: The center-bias weight α in the CPA module controls the weight ratio allocated to the central pixel in the center-biased average pooling strategy, where a higher α indicates a greater emphasis on the central region, and the remaining weight is equally assigned to surrounding context pixels. To verify the effectiveness of this mechanism, we adjusted α from 0.1 to 0.9 in Fig. 9 and observed that the OA consistently improves with the increase of α within the lower range, validating that enhancing the central feature representation can effectively suppress redundant interference. In particular, the OA increases steadily when α rises from 0.1 to 0.5, improving from 92.35% to 92.40% on Holden (HC), from 85.76% to 86.02% on NF (NF), and from 89.11% to 89.38% on Utopia (UP). Beyond $\alpha = 0.5$, the performance remains stable with only marginal fluctuations, indicating that the strategy maintains effectiveness across a wide range. Accordingly, $\alpha = 0.5$ is adopted as the default setting, as it achieves strong performance while maintaining a balanced contribution between central features and contextual information.

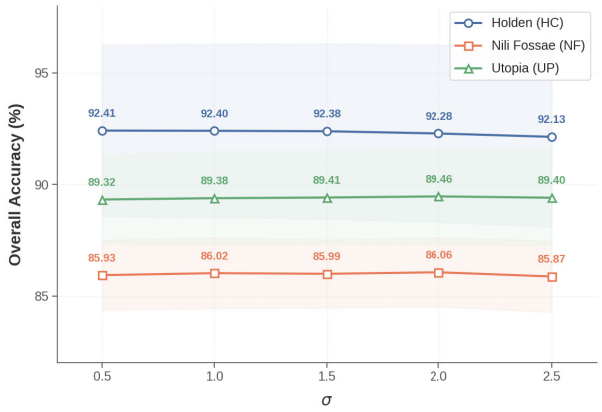


Fig. 10. OA with standard deviation on the HC, NF, and UP datasets for different Gaussian prompts σ.

5) Gaussian Prompt σ Sensitivity: The parameter σ controls the spatial extent of the Gaussian attention mask in the spatial Gaussian prompting branch. We adopted $\sigma = 1$ as the default configuration, which corresponds to the standard normal distribution. As shown in Fig. 10, the model exhibits strong robustness across different σ values. On the HC dataset, accuracy slightly degrades as σ increases, dropping from 92.41% at $\sigma = 0.5$ to 92.13% at $\sigma = 2.5$. The NF and UP datasets show similar stability with only marginal fluctuations. This trend aligns with the core design principle of the CPA module, where the spatial Gaussian prompting branch is intended to emphasize the central pixel. A smaller σ produces a sharper attention gradient that concentrates more weight on the target pixel while suppressing interference from surrounding context. The superior performance with a smaller σ validates the effectiveness of this center-focused strategy in reducing peripheral distractions and enhancing target discrimination.

D. Comparison With Other Methods

To further validate the superiority of the proposed framework under limited labeled samples, we conduct

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TABLE IV
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE HC DATASET

No.	GAHT	SSFTT	DSFormer	MASSFormer	APMN	OFD	SSEFN	FactoFormer	MCTGCL	MMFormer
1	96.53±4.91	97.68±3.65	95.90±4.96	96.44±5.86	96.02±5.22	96.59±5.91	94.98±6.07	95.68±6.11	95.44±6.03	**99.15±1.43**
2	84.79±9.56	85.74±8.14	84.71±8.63	86.67±8.88	85.41±8.33	85.29±9.54	85.64±10.48	82.52±9.54	88.80±9.34	**90.16±8.31**
3	86.93±7.43	83.97±7.01	80.90±6.31	84.99±7.30	84.81±6.80	88.18±8.34	86.50±6.50	81.03±9.76	87.04±6.28	**91.10±6.01**
4	77.26±7.89	77.78±6.46	69.28±5.89	80.70±6.94	75.80±7.98	80.29±6.18	73.71±4.71	75.60±5.53	81.39±7.16	**88.06±7.54**
5	90.78±5.21	88.12±6.16	86.80±7.85	91.01±5.20	87.28±8.48	91.31±5.70	87.80±6.57	92.55±9.37	93.71±5.79	**95.43±4.35**
6	99.46±0.72	99.24±0.90	98.88±1.42	98.75±1.25	98.95±1.13	**99.53±0.75**	99.00±1.27	96.74±4.79	98.93±1.30	98.95±1.09
OA	85.81±3.27	84.73±3.81	80.80±3.84	87.07±3.48	83.63±4.84	87.23±3.22	83.23±3.34	85.28±5.56	88.75±4.02	**92.40±3.89**
AA	89.29±2.36	88.76±2.24	86.08±2.13	89.76±2.56	88.05±2.55	90.20±2.07	87.94±2.21	87.35±3.02	90.88±2.80	**93.81±2.41**
Kappa	79.74±4.64	78.34±5.11	73.06±5.17	81.51±4.81	77.00±6.38	81.81±4.37	76.37±4.40	79.11±7.47	83.90±5.52	**89.05±5.47**
FLOPs (G)	0.5358	0.0456	0.1205	0.1349	0.0902	0.0422	0.5071	0.0710	0.1136	0.8591
Params (M)	1.0576	0.1478	0.5930	0.4087	0.5326	0.2497	9.8305	0.1429	0.2222	4.4358

TABLE V
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE NF DATASET

No.	GAHT	SSFTT	DSFormer	MASSFormer	APMN	OFD	SSEFN	FactoFormer	MCTGCL	MMFormer
1	84.13 ± 6.57	80.26 ± 11.10	83.58 ± 6.07	84.56 ± 7.42	87.96 ± 2.65	86.09 ± 7.13	78.99 ± 6.58	**94.53 ± 3.41**	88.88 ± 5.38	92.42 ± 3.25
2	84.89 ± 11.33	84.00 ± 7.04	83.16 ± 11.15	85.59 ± 10.52	84.85 ± 9.70	83.94 ± 13.77	**89.04 ± 10.40**	84.82 ± 9.41	80.34 ± 13.18	81.97 ± 10.71
3	87.37 ± 5.75	87.45 ± 5.66	87.90 ± 8.42	89.44 ± 4.91	87.50 ± 9.10	89.07 ± 5.78	89.89 ± 7.20	**91.84 ± 4.64**	89.68 ± 3.83	88.40 ± 4.94
4	76.98 ± 5.09	71.34 ± 5.77	67.19 ± 10.61	72.45 ± 8.72	74.55 ± 8.30	76.04 ± 10.50	73.50 ± 7.73	60.19 ± 21.68	77.45 ± 6.62	**86.04 ± 5.55**
5	69.45 ± 13.45	65.78 ± 10.38	58.94 ± 12.81	69.22 ± 12.99	64.08 ± 13.66	66.61 ± 16.84	67.61 ± 16.00	53.53 ± 14.66	75.46 ± 11.21	**78.67 ± 13.72**
6	66.64 ± 7.61	59.82 ± 8.51	59.80 ± 7.58	63.70 ± 7.29	64.31 ± 6.26	66.88 ± 6.63	62.91 ± 8.18	62.16 ± 10.99	70.81 ± 8.45	**75.34 ± 4.15**
7	78.83 ± 9.28	80.47 ± 8.81	78.47 ± 7.48	82.50 ± 7.42	78.17 ± 4.61	81.79 ± 7.45	81.41 ± 10.30	69.66 ± 10.46	81.07 ± 8.59	**85.52 ± 9.59**
8	85.94 ± 1.98	83.80 ± 2.06	82.01 ± 3.99	84.98 ± 3.85	85.79 ± 1.77	84.98 ± 3.71	85.05 ± 4.04	81.95 ± 8.05	**86.16 ± 3.22**	85.99 ± 4.07
9	78.35 ± 6.99	71.05 ± 5.68	72.00 ± 9.06	71.99 ± 7.19	79.36 ± 8.67	77.17 ± 6.86	72.33 ± 5.28	84.05 ± 10.87	75.56 ± 10.89	**86.41 ± 3.33**
OA	78.54 ± 2.57	73.58 ± 4.53	74.12 ± 3.61	76.27 ± 4.51	79.21 ± 2.29	78.87 ± 3.14	74.63 ± 2.40	79.72 ± 3.93	80.26 ± 4.09	**86.02 ± 1.59**
AA	79.18 ± 1.90	75.99 ± 2.68	74.78 ± 2.04	78.27 ± 3.62	78.51 ± 1.88	79.17 ± 2.97	77.86 ± 2.40	75.86 ± 3.77	80.60 ± 2.53	**84.53 ± 3.35**
Kappa	73.35 ± 3.09	67.32 ± 5.40	67.90 ± 4.27	70.54 ± 5.53	74.06 ± 2.81	73.88 ± 3.73	68.77 ± 2.82	74.71 ± 4.70	75.37 ± 4.87	**82.51 ± 1.93**
FLOPs (G)	0.5358	0.0456	0.1205	0.1349	0.0902	0.0422	0.5072	0.0710	0.1136	0.8591
Params (M)	1.0578	0.1480	0.5930	0.4091	0.5330	0.2501	9.8421	0.1431	0.2225	4.4360

TABLE VI
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE UP DATASET

No.	GAHT	SSFTT	DSFormer	MASSFormer	APMN	OFD	SSEFN	FactoFormer	MCTGCL	MMFormer
1	70.98 ± 9.86	66.67 ± 8.60	57.55 ± 10.91	68.58 ± 9.33	63.22 ± 6.77	68.76 ± 6.70	59.36 ± 7.76	60.28 ± 11.11	71.73 ± 9.31	**81.18 ± 12.55**
2	84.23 ± 6.48	83.09 ± 7.64	78.78 ± 8.92	83.27 ± 9.76	83.09 ± 6.00	84.31 ± 6.94	82.89 ± 6.74	77.59 ± 6.74	83.46 ± 7.10	**84.78 ± 6.48**
3	92.01 ± 4.79	89.95 ± 7.83	85.85 ± 5.83	92.96 ± 7.04	89.76 ± 8.19	91.85 ± 7.27	94.55 ± 5.17	80.85 ± 13.30	91.00 ± 5.99	**97.44 ± 2.75**
4	91.22 ± 5.80	91.87 ± 5.73	92.52 ± 6.09	93.78 ± 7.21	89.54 ± 10.05	90.53 ± 10.80	**96.37 ± 6.04**	89.08 ± 12.03	93.09 ± 3.62	93.59 ± 4.47
5	89.79 ± 4.73	91.53 ± 6.06	89.09 ± 5.55	93.82 ± 5.39	90.69 ± 5.61	90.40 ± 5.92	90.06 ± 4.72	83.86 ± 7.35	**94.73 ± 4.70**	93.33 ± 5.68
6	95.44 ± 5.16	94.52 ± 6.88	95.00 ± 4.15	94.29 ± 5.49	95.74 ± 2.92	**96.33 ± 3.88**	95.76 ± 4.18	89.00 ± 8.05	93.43 ± 6.04	94.90 ± 5.07
7	99.21 ± 2.49	97.27 ± 8.62	99.85 ± 0.48	98.30 ± 5.37	99.27 ± 2.30	98.30 ± 5.37	**100.00 ± 0.00**	98.27 ± 2.85	**100.00 ± 0.00**	99.88 ± 0.38
8	92.45 ± 4.27	84.69 ± 8.76	86.41 ± 7.72	87.66 ± 8.35	95.94 ± 3.39	**97.50 ± 3.37**	92.03 ± 4.72	83.18 ± 15.22	90.62 ± 6.04	97.24 ± 2.33
9	81.16 ± 5.59	77.64 ± 8.00	77.17 ± 7.12	80.26 ± 7.03	79.95 ± 6.07	79.85 ± 6.30	78.72 ± 7.86	63.27 ± 9.12	82.45 ± 7.49	**86.70 ± 7.73**
OA	86.39 ± 2.43	85.91 ± 2.84	82.58 ± 3.38	87.29 ± 3.70	85.52 ± 2.56	86.47 ± 2.46	84.97 ± 2.29	79.29 ± 3.05	88.06 ± 2.48	**89.38 ± 2.09**
AA	88.50 ± 1.92	86.36 ± 2.96	84.69 ± 2.24	88.10 ± 2.77	87.47 ± 1.42	88.65 ± 2.30	87.75 ± 1.96	80.60 ± 2.41	88.94 ± 1.59	**92.12 ± 1.20**
Kappa	81.35 ± 3.25	80.70 ± 3.80	76.45 ± 4.29	82.61 ± 4.95	80.22 ± 3.30	81.42 ± 3.25	79.51 ± 3.05	71.59 ± 3.89	83.54 ± 3.32	**85.33 ± 2.84**
FLOPs (G)	0.5358	0.0450	0.1205	0.1349	0.0902	0.0422	0.5072	0.0710	0.1136	0.8591
Params (M)	1.0578	0.1480	0.5930	0.4091	0.5330	0.2501	9.8421	0.1431	0.2225	4.4360

comprehensive comparative experiments with nine representative state-of-the-art HSIC methods. Among them, several approaches are specifically designed for few-shot learning and small-sample regimes, while others are representative mainstream HSI classification models to ensure comprehensive comparison. All methods are evaluated on three benchmark Mars HSI datasets, HC, NF, and UP, under identical experimental configurations, including ten labeled samples per class and ten independent runs, to ensure statistical reliability. Detailed quantitative results are reported in Tables IV–VI, where the top two performances are marked in bold and underlined, respectively. Corresponding classification maps are visualized in Figs. 11–13 for intuitive performance analysis. Brief descriptions of each comparison approach are given as follows.

1) *GAHT [40]*: A hierarchical Transformer employing grouped pixel embedding (GPE) to constrain MHSA within local spatial–spectral contexts, addressing feature over-dispersion in ViTs and enabling high accuracy with few parameters.
2) *SSFTT [43]*: A spectral–spatial feature tokenization Transformer that uses 3-D/2-D convolutions for shallow feature extraction and a Gaussian-weighted tokenizer,

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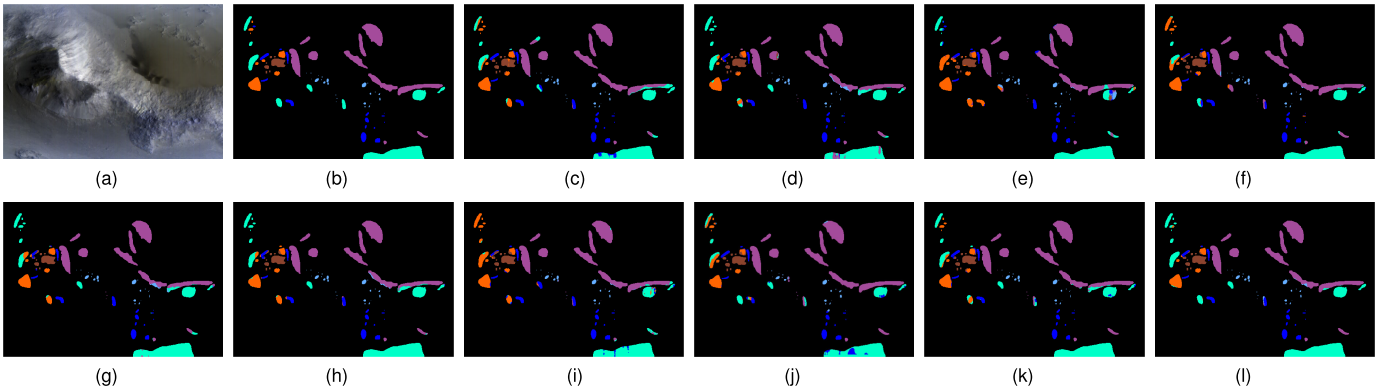


Fig. 11. Classification maps obtained via various algorithms for the HC dataset. (a) False color. (b) Ground truth. (c) GAHT. (d) SSFTT. (e) DSFormer. (f) MASSFormer. (g) APMN. (h) OFD. (i) SSEFN. (j) FactoFormer. (k) MCTGCL. (l) MMFormer.

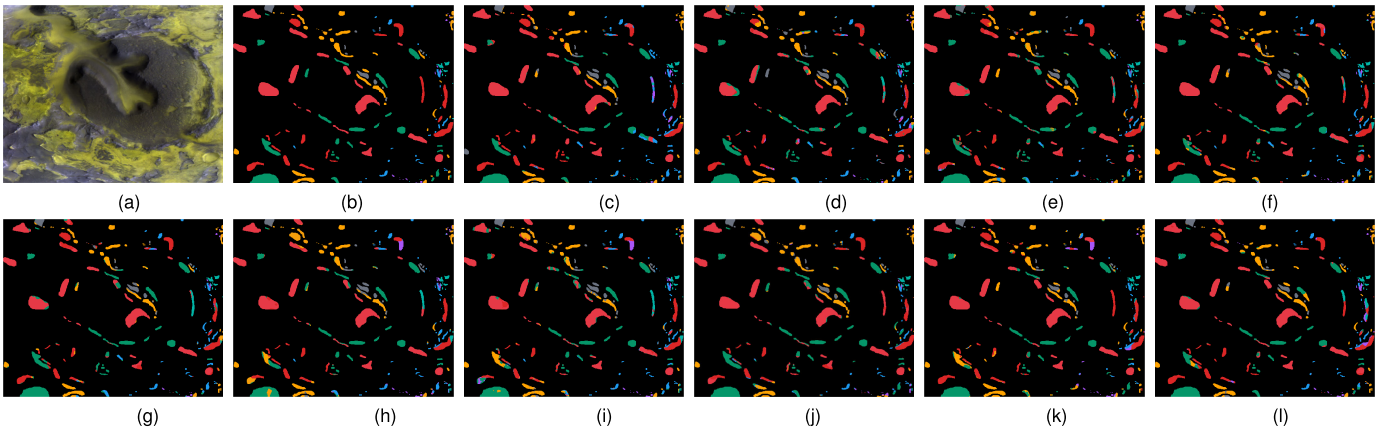


Fig. 12. Classification maps obtained via various algorithms for the NF dataset. (a) False color. (b) Ground truth. (c) GAHT. (d) SSFTT. (e) DSFormer. (f) MASSFormer. (g) APMN. (h) OFD. (i) SSEFN. (j) FactoFormer. (k) MCTGCL. (l) MMFormer.

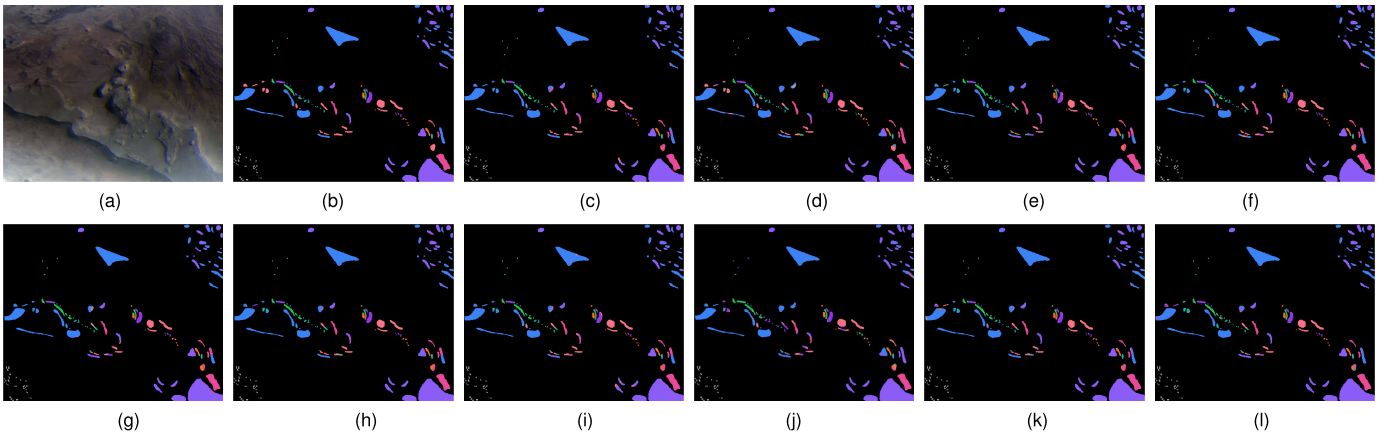


Fig. 13. Classification maps obtained via various algorithms for the UP dataset. (a) False color. (b) Ground truth. (c) GAHT. (d) SSFTT. (e) DSFormer. (f) MASSFormer. (g) APMN. (h) OFD. (i) SSEFN. (j) FactoFormer. (k) MCTGCL. (l) MMFormer.

achieving efficient high-level semantic representation with low computational cost.

3) *DSFormer [32]*: A dual selective fusion network that introduces kernel selective fusion (KSFTB) and token selective fusion (TSFTB) blocks to adaptively fuse multiscale spatial–spectral features, overcoming the limitations of fixed receptive fields in standard Transformers.
4) *MASSFormer [41]*: A memory-augmented spectral–spatial Transformer that utilizes a memory tokenizer (MT) and memory-augmented Transformer encoder (MATE) modules to integrate prior knowledge, enhancing spectral–spatial mixing and mitigating depth decay.
5) *APMN [62]*: An attention-based patch mixing network designed for few-shot learning, which synthesizes complex training samples via online patch mixing and

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employs pixel-to-class attention to enhance robustness against sample scarcity.

6) *OFD [63]*: An orthogonal feature disentanglement network for cross-domain few-shot classification. It projects features into domain-shared and domain-specific subspaces via orthogonal decomposition, effectively suppressing domain shift interference with limited target-domain labels.
7) *SSEFN [64]*: A spatial–spectral enhancement network tailored for small-sample regimes. It utilizes an adaptive decision fusion (ADF) module to combine predictions from enhanced features, effectively alleviating overfitting caused by insufficient labeled data.
8) *FactoFormer [65]*: A factorized spectral–spatial Transformer with self-supervised pretraining. It decouples spectral and spatial modeling and employs masked image modeling (MIM) strategies to address the data-hungry nature of Transformers in HSI tasks.
9) *MCTGCL [1]*: A mixed CNN–Transformer framework for Mars HSI classification. It adopts a lightweight dual-branch structure and graph contrastive learning to efficiently extract hybrid local–global features under data-limited scenarios.

1) Analysis on the HC Dataset: The experimental results on the HC dataset, summarized in Table IV, indicate that MMFormer achieves the best performance among all comparison methods. In particular, MMFormer attains an OA of 92.40%, surpassing the second-best method, MCTGCL, by 3.65%. Notably, our method also secures the highest AA of 93.81% and Kappa coefficient of 89.05%, which suggests a more balanced and stable classification capability across all categories compared to other models. Compared with competitive methods, including FactoFormer and MASSFormer, our model obtains prominent performance advantages. Although these approaches construct effective spectral and spatial dependency modeling pipelines, their generalization capability tends to be limited under small labeled sample conditions, leading to insufficient feature discrimination for noisy and ambiguous pixels. By contrast, MMFormer shows superior capacity in capturing fine-grained spatial–spectral details. For instance, on Class 1 (Analcime), which requires precise discrimination of geological boundaries, MMFormer achieves a remarkable accuracy of 99.15%. This is significantly higher than FactoFormer (95.68%) and MASSFormer (96.44%), validating that our architecture effectively preserves the integrity of dominant geological components while filtering out spectral interference.

Furthermore, MMFormer exhibits exceptional robustness in handling classes with extremely scarce samples, a critical challenge in Mars hyperspectral classification. Benefiting from the elaborate model design, MMFormer achieves superior generalization ability from limited training data compared with other competing algorithms. This advantage is most evident in Class 6 (Margarite), which contains only a small number of available samples. Most comparative methods encounter varying degrees of performance limitations, such as MASSFormer (98.75%) and DSFormer (98.88%). Nevertheless, MMFormer maintains a high accuracy of 98.95%. Such robust classification results on both dominant and rare classes verify that the proposed macro–micro kernel strategy and CPA module can effectively extract discriminative spatial–spectral features under strict data constraints, and guarantee stable classification performance against inherent sample imbalance in Martian geological datasets.

2) Analysis on the NF Dataset: The experimental results on the NF dataset are presented in Table V. Unlike the HC dataset, this region suffers from severe spectral aliasing, intricate mineral distribution, and ambiguous category boundaries, leading to generally unsatisfactory classification performance across almost all competing methods. Under such challenging classification scenarios, MMFormer still achieves the optimal overall performance, with an OA of 86.02%, which remarkably surpasses the second-best method MCTGCL by 5.76%. Meanwhile, our method also obtains the highest AA of 84.53% and Kappa coefficient of 82.51% among all compared approaches, demonstrating outstanding overall superiority and stable classification performance.

Compared with other competitive models, most methods encounter obvious performance limitations when facing highly mixed spectral information and complex geological environments. Their extracted spatial–spectral features are insufficiently discriminative, resulting in notable accuracy degradation on multiple confusing mineral categories. On the contrary, MMFormer exhibits prominent advantages in distinguishing spectrally similar components. Benefiting from the elaborate model architecture, the proposed method maintains excellent feature extraction capability even under severe spectral interference. In particular, MMFormer secures the first place in 5 out of 9 categories, and achieves remarkable accuracy advantages on multiple classes with serious spectral confusion. Such results verify that the designed module structures can effectively capture comprehensive spatial–spectral correlation information and suppress redundant spectral interference in complex Martian geological environments.

According to the classification maps displayed in Fig. 12, the prediction result of MMFormer shows satisfactory regional structural integrity and continuous boundary distribution. Compared with other methods that present massive scattered noise and misclassified pixels, our model effectively retains fine-grained geological details and maintains consistent regional classification results, further demonstrating its robust generalization performance in the most challenging Mars hyperspectral classification task.

3) Analysis on the UP Dataset: The experimental results on the UP dataset are summarized in Table VI. Characterized by fine-grained geological textures and subtle spectral differences between similar minerals, this region poses significant challenges for feature discrimination, leading to tight performance competition among all comparison methods. Despite these stringent conditions, MMFormer consistently achieves the optimal overall performance, with an OA of 89.38%, surpassing the second-best method MCTGCL by 1.32%. More significantly, our method secures the highest AA of 92.12% and Kappa coefficient of 85.33%, demonstrating exceptional classification stability across all categories.

Benefiting from the robust dual-branch architecture, MMFormer presents prominent advantages in capturing

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TABLE VII
CLASSIFICATION PERFORMANCE (%) OF THE ABLATIONS ON THE PROPOSED METHOD

Data set	Case	Spectral Former	Spatial Former	CPA	CWF	OA (%)	AA (%)	Kappa (%)	Params (M)	FLOPs (G)
HC	1	-	-	-	-	84.84 ± 3.24	87.74 ± 2.14	78.45 ± 4.42	0.6064	0.1008
	2	✓	-	-	-	89.75 ± 2.99	91.89 ± 1.86	85.36 ± 4.09	1.5975	0.3809
	3	✓	✓	-	-	90.84 ± 2.89	92.73 ± 2.12	86.84 ± 4.03	3.8099	0.7554
	4	✓	✓	✓	-	91.09 ± 3.69	93.10 ± 2.15	87.22 ± 5.11	4.0188	0.7886
	5	✓	✓	✓	✓	$\mathbf{92.40 \pm 3.89}$	$\mathbf{93.81 \pm 2.41}$	$\mathbf{89.05 \pm 5.47}$	4.4358	0.8591
NF	1	-	-	-	-	73.38 ± 5.42	72.22 ± 2.98	67.08 ± 6.18	0.6066	0.1008
	2	✓	-	-	-	80.48 ± 3.55	80.95 ± 2.71	75.82 ± 4.23	1.5977	0.3809
	3	✓	✓	-	-	83.17 ± 2.19	82.10 ± 2.16	79.01 ± 2.65	3.8101	0.7554
	4	✓	✓	✓	-	85.28 ± 2.45	84.34 ± 3.61	81.62 ± 3.01	4.0190	0.7886
	5	✓	✓	✓	✓	$\mathbf{86.02 \pm 1.59}$	$\mathbf{84.53 \pm 3.35}$	$\mathbf{82.51 \pm 1.93}$	4.4360	0.8591
UP	1	-	-	-	-	83.08 ± 3.93	83.03 ± 3.32	76.68 ± 5.28	0.6066	0.1008
	2	✓	-	-	-	87.65 ± 2.33	90.04 ± 1.31	83.05 ± 3.13	1.5977	0.3809
	3	✓	✓	-	-	88.67 ± 1.76	90.63 ± 1.24	84.38 ± 2.37	3.8101	0.7554
	4	✓	✓	✓	-	89.06 ± 1.97	91.88 ± 1.48	84.88 ± 2.68	4.0190	0.7886
	5	✓	✓	✓	✓	$\mathbf{89.38 \pm 2.09}$	$\mathbf{92.12 \pm 1.20}$	$\mathbf{85.33 \pm 2.84}$	4.4360	0.8591

comprehensive spatial and spectral feature information under complex fine-grained scenes. The model maintains excellent feature discriminative capability for minerals with weak spectral differences and achieves leading performance on multiple challenging categories. For instance, in Class 3 with highly similar spectral signatures, MMFormer reaches an impressive accuracy of 97.44%, remarkably exceeding competing methods such as SSEFN (94.55%).

The embedded attention module and subsequent feature fusion strategy further support stable feature representation. In particular, MMFormer ranks first in 4 out of 9 categories, showing robust superiority across diverse fine-grained mineral classes. As illustrated in Fig. 13, the classification map generated by MMFormer is highly consistent with the ground truth, exhibiting sharper class boundaries, more uniform regional responses, and stronger robustness to spectral disturbances on the Martian surface. The comprehensive experimental results fully verify the excellent adaptability of the proposed framework for fine-grained Mars hyperspectral classification tasks.

4) Model Complexity and Efficiency Analysis: Model complexity is further analyzed in terms of floating-point operations and learnable parameters. The corresponding statistical results are presented in Tables IV–VI. MMFormer has a computational cost of 0.8591 GFLOPs and about 4.4360M parameters. These values are relatively higher than those of some lightweight competitors due to the dual-branch structure and multiscale attention design. Overall, the model maintains a moderate parameter scale among mainstream hyperspectral classification models in practical deep-learning applications. The additional complexity brings clear performance gains, especially in small-sample Mars mineral classification tasks where modeling capacity is crucial. Compared with the second-best method, the proposed model achieves consistent and substantial improvements across all three datasets. In particular, our method obtains 3.65% higher OA on the HC dataset, 5.76% higher OA on the NF dataset, and 1.32% higher OA on the UP dataset. The proposed model achieves excellent classification performance with acceptable computational overhead.

E. Ablation Experiments

Ablation studies are conducted on the HC, NF, and UP datasets to validate the contribution of each proposed component. The results are presented in Table VII. All configurations share the same shallow feature extraction block and classification head. The five experimental cases are designed progressively as follows.

1) *Case 1 (Baseline)*: Uses three standard PoolFormer blocks as the feature extractor.
2) *Case 2*: Replaces PoolFormer with three spectral MMFormer blocks (CPA disabled).
3) *Case 3*: Adds a parallel spatial MMFormer branch to Case 2 (dual-branch, no CPA).
4) *Case 4*: Integrates the CPA module into both branches of Case 3.
5) *Case 5*: Adds the CWF module to Case 4 for further feature fusion.

The experimental results demonstrate a clear and progressive performance improvement across all datasets. The baseline (Case 1) serves as a reference point, while replacing its PoolFormer blocks with the proposed spectral MMFormer (Case 2) achieves substantial OA improvements of 4.91%, 7.10%, and 4.57% on the three datasets. This fully verifies that the spectral MMFormer can efficiently capture the rich spectral information of HSI and mine hierarchical spectral correlations to enhance feature discriminability.

Introducing the spatial MMFormer branch to form a dual-branch architecture (Case 3) further delivers consistent OA gains of 1.09%, 2.69%, and 1.02%, confirming that the spatial MMFormer can effectively extract multiscale spatial context information of HSI, while the dual-branch structure realizes complementary fusion between spatial and spectral features to enrich feature representation. Integrating the CPA module into both branches (Case 4) achieves additional meaningful OA enhancements of 0.25%, 2.11%, and 0.39%, indicating that the CPA module can effectively emphasize the central pixel features and suppress background interference, thus further improving the accuracy of pixel-level classification. Finally, adding the CWF module for cross-branch fusion (Case 5)

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attains the optimal performance, with final OA increments of 1.31%, 0.74%, and 0.32% over Case 4. This demonstrates that the CWF module can adaptively integrate the complementary information of dual branches, fully leveraging the advantages of spatial and spectral feature extraction and further improving the overall classification performance.

IV. Conclusion and Future Work

A. Conclusion

This study proposes the MMFormer framework to address the challenges of high spectral similarity, scarce labeled samples, and environmental interference in Mars HSIC. In particular, the framework adopts a dual-branch architecture composed of spatial MMFormer and spectral MMFormer modules. With the *macro-kernel perception and micro-kernel modulation* strategy, the dual-branch structure effectively captures multiscale spatial contextual dependencies and comprehensive spectral band correlations, enabling comprehensive and discriminative feature representation. For further performance optimization, the embedded CPA module enhances the discriminative representation of central target pixels. The subsequent CWF module adaptively integrates complementary spatial and spectral information. Comprehensive experimental results across three hyperspectral datasets demonstrate that MMFormer achieves consistent and superior performance against state-of-the-art methods. The overall results verify the effectiveness and adaptability of the proposed framework, demonstrating its strong applicability to Mars hyperspectral classification tasks with limited labeled samples and complex geological conditions.

B. Future Work

Although MMFormer has demonstrated promising performance, several directions remain for future research to enhance its applicability as follows.

1) *Unsupervised Learning*: Current methods rely heavily on labeled data. Future work will explore unsupervised or self-supervised learning strategies to leverage the vast amounts of unlabeled data acquired by orbiters, thereby reducing dependence on manual annotations.
2) *Onboard Efficiency*: Real-time analysis on spacecraft demands high computational efficiency. We plan to investigate model compression techniques, such as quantization and pruning, to facilitate the deployment of MMFormer on resource-constrained hardware like Mars rovers.
3) *Cross-Domain Generalization*: Spectral characteristics can vary significantly across different Martian regions. Developing domain adaptation algorithms will be essential to enable the model to generalize across diverse geological terrains without the need for retraining.

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